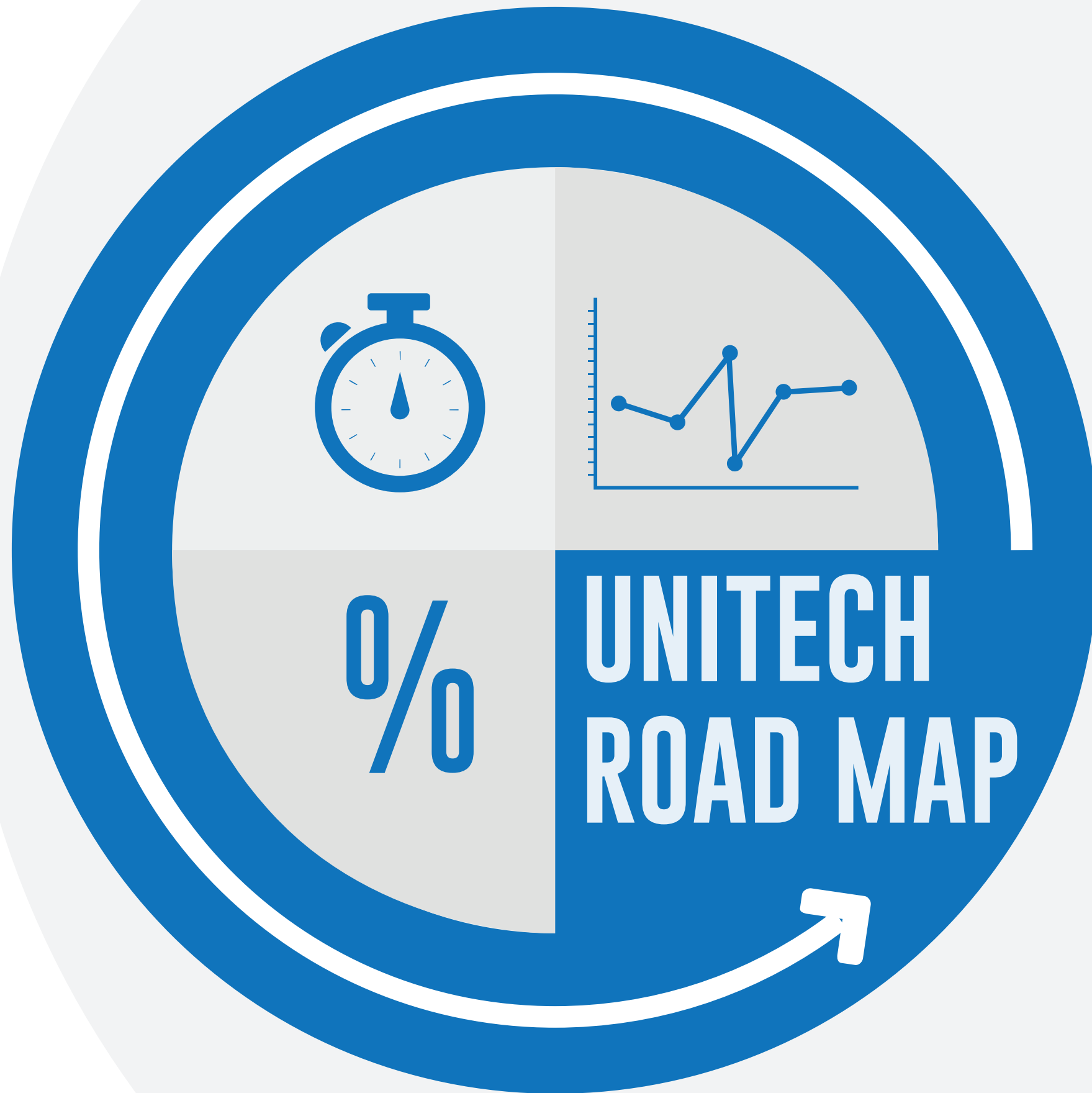




Challenge !
The times to come

COST OF QUALITY
(COQ)

COST OF QUALITY

INTRODUCTION

THE COST OF QUALITY IS A WIDELY USED TERM AND WIDELY MISUNDERSTOOD. THE “COST OF QUALITY” IS NOT THE PRICE OF CREATING A QUALITY PRODUCT OR SERVICE. ITS THE COST OF **NOT** CREATING A QUALITY PRODUCT OR SERVICE THE **FIRST TIME, EVERY TIME.**

QUALITY COSTS CAN BE CATEGORIZED TO ENABLE BETTER UNDERSTANDING:

- PREVENTION COST
- APPRAISAL COST
- INTERNAL FAILURE COSTS
- EXTERNAL FAILURE COST
- INDIRECT QUALITY COST

VISIBLE / HIDDEN COST OF QUALITY:

TAKE A LOOK AT THE EXAMPLE OF THE ICEBERG HIGHLIGHTING THE HIDDEN COST OF POOR QUALITY.



COST OF QUALITY

COST DRIVER ANALYSIS

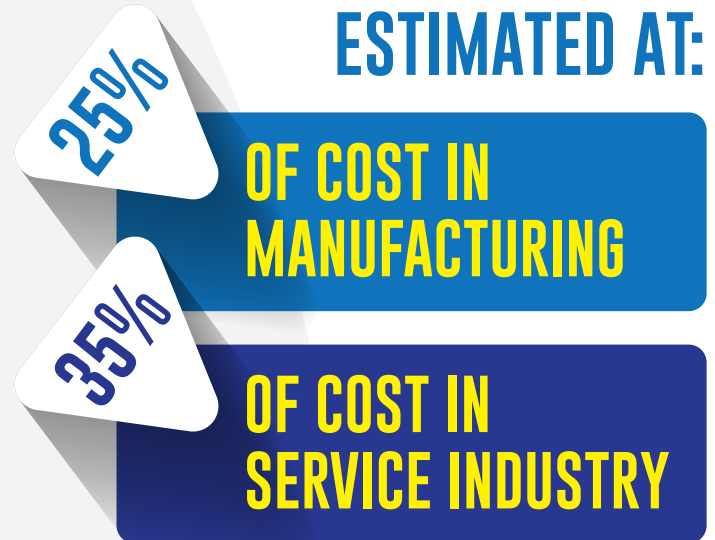
A POWERFUL TECHNIQUE THAT INTEGRATES THE PROBLEM SOLVING TOOLS OF THE QUALITY PROCESS WITH THE POOR-QUALITY COST ASSESSMENT.

HOW IS COST DRIVER ANALYSIS PERFORMED?

QUALITY COSTS CAN BE BETTER UNDERSTOOD BY IMPLEMENTING THE FOLLOWING STEPS:

- 1 DETERMINE THE ROOT CAUSE OF THE POOR QUALITY COSTS
- 2 IDENTIFY THE ACTIVITY PERCENTAGES AND CALCULATE THE COST OF POOR QUALITY RELATED TO EACH ROOT CAUSE
- 3 COMBINE THE FINANCIAL IMPACTS OF COMMON ROOT CAUSES TO DETERMINE THE TOTAL FINANCIAL IMPACT OF THE ROOT CAUSE
- 4 PERFORM A COST-BENEFIT ANALYSIS TO DETERMINE THE FINANCIAL IMPACT

COST OF QUALITY IS ESTIMATED AT:



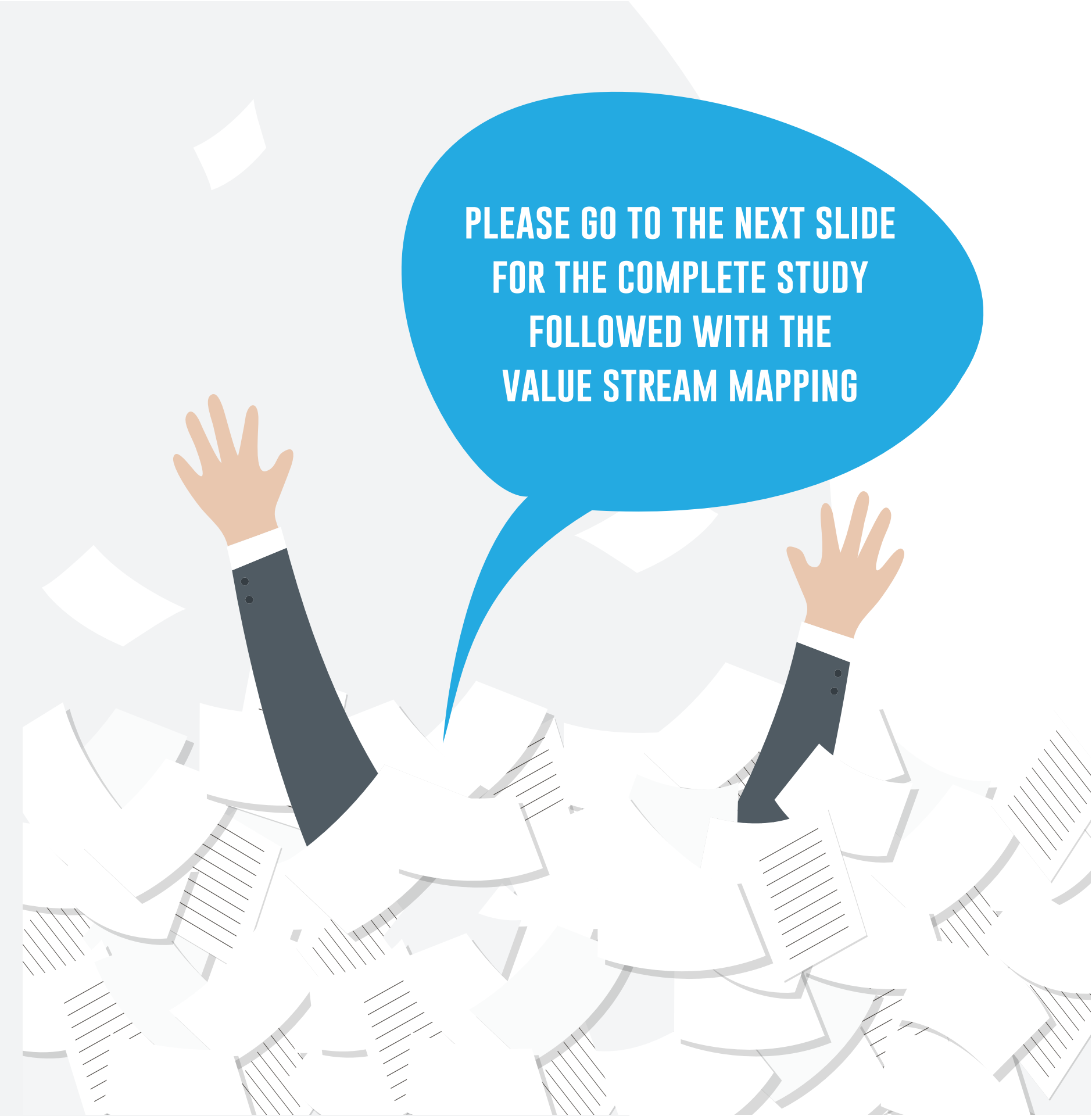
COST OF QUALITY

THE STUDY

DURING 2015, A STUDY WAS CARRIED OUT IN **SFSP KSA** ON A **CABLE TRAY MANUFACTURING LINE** TO DETERMINE THE COST OF QUALITY IN ORDER TO FURTHER IMPROVE THE CURRENT STATE OF PRODUCTION BY REDUCING OR ELIMINATING WASTE AND NON-VALUE ADDED ACTIVITIES FROM OUR VARIOUS PRODUCTION PROCESS.

WHILE WE WERE CONDUCTING THE COST DRIVER ANALYSIS, WE WERE ANALYZING EVERY COST, AND COST CATEGORY, AND ASKING THIS QUESTION:

“IF ALL OF OUR PROCESSES PRODUCED THE CORRECT RESULTS THE FIRST TIME, WOULD THIS COST STILL BE HERE??”



PLEASE GO TO THE NEXT SLIDE
FOR THE COMPLETE STUDY
FOLLOWED WITH THE
VALUE STREAM MAPPING

COST OF QUALITY – THE STUDY

COST OF QUALITY ELEMENTS

REF:	DESCRIPTION	UNIT	AMOUNT	INFORMATION / SOURCE RECORD
Q1	TIME SPENT ON EMPLOYEE TRAINING (OUTSOURCED ACTIVITY)	HOURS	4.00	EMPLOYEE TRAINING
Q2	TIME SPENT ON RAW MATERIAL INSPECTION (MANUAL ACTIVITY)	HOURS	9.00	ACTIVITIES PER EMPLOYEE
Q3	TIME SPENT ON FINAL PRODUCT INSPECTION (MANUAL ACTIVITY)	HOURS	208.00	ACTIVITIES PER EMPLOYEE
Q4	LABOUR COST (PER HOUR) OF RAW MATERIAL INSPECTION	SAR	24.00	LABOR COST
Q5	LABOUR COST (PER HOUR) OF FINAL PRODUCT INSPECTION	SAR	24.00	LABOR COST
Q6	LABOUR COST (PER HOUR) OF PRODUCTION	SAR	16.00	LABOR COST
Q7	TIME SPENT ON PREVENTIVE MAINTENANCE (OUTSOURCED ACTIVITY)	HOURS	60.00	PREVENTIVE MAINTENANCE
Q8	AVERAGE PRICE PER PRODUCT UNIT SOLD	SAR	120.00	PRICE LIST / CUSTOMER CONTRACT
Q9	COMPENSATION COST PAID TO CUSTOMERS PER FAILED PRODUCT	SAR	35.00	CUSTOMER CONTRACT
Q10	PRODUCT UNITS REJECTED BY INTERNAL INSPECTIONS (INTERNAL FAILURES)	UNITS	2,000	NON-CONFORMING
Q11	PRODUCT UNITS REJECTED BY CUSTOMERS (EXTERNAL FAILURES)	UNITS	1,200	NON-CONFORMING
Q12	RAW MATERIALS CONSUMED PER PRODUCT UNIT	KGS	16.50	PRODUCT DESIGN
Q13	LABOUR CONSUMED PER PRODUCT UNIT	HOURS	0.600	PRODUCT DESIGN
Q14	COST (PER HOUR) EMPLOYEE TRAINING (FEES PAID TO TRAINERS)	SAR	250.00	OUTSOURCED SERVICE COST
Q15	COST (PER HOUR) OF PREVENTIVE MAINTENANCE	SAR	120.00	OUTSOURCED SERVICE COST
Q16	COST (PER KG) OF RAW MATERIAL	SAR	3.70	RAW MATERIAL COST
A17	AVERAGE COST PER PRODUCT UNIT	SAR	61.00	PRODUCT COSTING
Q18	AVERAGE NUMBER OF NON-SOLD UNITS PER EACH EXTERNAL FAILURE	UNITS	3	CUSTOMER SATISFACTION SURVEYS / SCORE CARDS
Q19	RAW MATERIAL TOTAL COST (LAST PERIOD)	SAR	1,282,000	INCOME STATEMENT
Q20	LABOUR TOTAL COST (LAST PERIOD) - (SALARIES/WAGES)	SAR	220,000	INCOME STATEMENT
Q21	OUTSOURCED SERVICES TOTAL COST (LAST PERIOD)	SAR	35,000	INCOME STATEMENT
Q22	OTHER EXPENSES (LAST PERIOD)	SAR	76,000	INCOME STATEMENT

COST OF QUALITY – THE STUDY

THE DETAILED COST OF QUALITY

		FINANCIAL ACCOUNTS					
REF. N.	COST OF QUALITY SUB - CATEGORY	RAW MATERIAL (A)	SALARIES/ WAGES (B)	OUTSOURCED SERVICE (C)	OTHER EXPENSES (D)	INDIRECT COST (E)	TOTAL
1.1	EMPLOYEE TRAINING COST	0	0	1,000	0	0	1,000
1.2	PREVENTIVE MAINTENANCE COST	0	0	7,200	0	0	7,200
1.3	RAW MATERIAL INSPECTION COST	0	216	0	0	0	216
1.4	FINAL PRODUCT INSPECTION COST	0	4,992	0	0	0	4,992
1.5	COST OF PRODUCTS REJECTED INTERNALLY	122,100	19,200	0	0	0	141,300
1.6	COST OF PRODUCTS RETURNED BY CUSTOMERS	73,260	11,520	0	42,000	0	126,780
1.7	EARNING FORGONE DUE TO LOST SALES	0	0	0	0	212,400	212,400
TOTAL COST		195,360	35,928	8,200	42,000	212,400	493,888

EXPLAINING THE DETAILED COST OF QUALITY

1.1 EMPLOYEE TRAINING COST

THE ONLY FINANCIAL ACCOUNT AFFECTED BY EMPLOYEE TRAINING IS "OUTSOURCED SERVICES (C)" GIVEN THAT IT IS ASSUMED THAT THIS ACTIVITY IS PERFORMED BY EXTERNAL TEACHERS / TRAINERS. FEES PAID TO TRAINERS MAY BE CALCULATED BY MULTIPLYING THE TIME SPENT AND THE COST PER HOUR OF THE EMPLOYEE TRAINING. Q1 X Q14

1.2 PREVENTIVE MAINTAINANCE COST

THE ONLY FINANCIAL ACCOUNT MODIFIED BY PREVENTIVE MAINTENANCE IS ALSO "OUTSOURCED SERVICES (C)" BECAUSE IT IS ASSUMED THAT THIS ACTIVITY IS CARRIED OUT BY EXTERNAL SUPPLIERS. THE TOTAL COST FOR THIS ACCOUNT MAY BE CALCULATED BY MULTIPLYING THE TIME SPEND ON PREVENTIVE MAINTENANCE BY ITS COST PER HOUR. Q7 X Q15

COST OF QUALITY – THE STUDY

EXPLAINING THE DETAILED COST OF QUALITY (CONT.)

2.1 RAW MATERIAL INSPECTION COST

GIVEN THAT THE INSPECTION OF RAW MATERIALS IS PERFORMED MANUALLY BY INTERNAL STAFF, IT ONLY GENERATES LABOR COST AND, THEREFORE, MODIFIES THE "SALARIES/WAGES (B)" FINANCIAL ACCOUNT. THIS CAN BE DERIVED BY MULTIPLYING THE TIME SPENT ON RAW MATERIAL INSPECTION WITH THE LABOR COST OF THIS ACTIVITY PER HOUR.

$Q2 \times Q4$

2.2 FINAL PRODUCT INSPECTION COST

THE INSPECTION OF FINAL PRODUCT HAS AN IMPACT SOLELY ON THE "SALARIES/WAGES (B)" ACCOUNT; GIVEN THAT, BEING A MANUAL ACTIVITY, IT ONLY GENERATES LABOUR COSTS. THE TOTAL COST FOR THIS FINANCIAL ACCOUNT MAYBE ESTIMATED BY MULTIPLYING THE TIME SPENT ON FINAL PRODUCTION INSPECTION BY ITS COST PER HOUR.

$Q3 \times Q5$

3.1 COST OF PRODUCTS REJECTED INTERNALLY

IT IS ASSUMED THAT EVERY PRODUCT UNIT REJECTED BY INTERNAL INSPECTION MUST BE DISCARDED AND THAT THE ONLY RESOURCES USED IN THE PRODUCTIVE PROCESS ARE RAW MATERIAL AND LABOR. THIS MODIFIES TWO ACCOUNTS "RAW MATERIALS (A)" & "SALARIES/WAGES (B)".

THE AMOUNT OF RAW MATERIAL WASTE CAN BE DERIVED BY MULTIPLYING PRODUCT UNITS REJECTED BY INTERNAL INSPECTION BY RAW MATERIAL CONSUMED PER PRODUCT UNIT ($Q10 \times Q12$). THIS SHOULD BE FURTHER MULTIPLIED WITH THE COST (PER KG) OF RAW MATERIAL TO GET THE COST OF RAW MATERIAL.

$(Q10 \times Q12) \times Q16$

THE SALARY COST IS DERIVED BY MULTIPLYING THE LABOUR CONSUMED PER PRODUCT UNIT BY PRODUCT UNITS REJECTED BY INTERNAL INSPECTION ($Q13 \times Q10$), THE RESULT OF IT SHOULD BE FURTHER MULTIPLIED BY THE LABOR COST (PER HOUR) OF PRODUCTION; THE RESULT SHOULD REFLECT UNDER "SALARIES/WAGES (B)".

$(Q13 \times Q10) \times Q6$

COST OF QUALITY – THE STUDY

EXPLAINING THE DETAILED COST OF QUALITY (CONT.)

4.1 COST OF PRODUCTS RETURNED BY CUSTOMERS

ANOTHER ASSUMPTION MADE IS THAT, WHEN AN EXTERNAL FAILURE HAPPENS, THE PRODUCT IS DISCARDED, THE PRICE IS REIMBURSED AND A COMPENSATION AMOUNT IS PAID TO THE CLIENT. THIS MODIFIES THREE ACCOUNTS "RAW MATERIALS (A)" , "SALARIES/WAGES (B)" & "OTHER EXPENSES (D)"

TO DERIVE THE COST OF RAW MATERIAL, WE SHOULD FIRST DETERMINE THE RAW MATERIAL WASTE. WE SHALL THEN MULTIPLY RAW MATERIAL CONSUMED PER PRODUCT UNIT WITH PRODUCT UNIT REJECTED BY CUSTOMERS AND FURTHER MULTIPLYING THE RESULT WITH THE COST (PER KG) OF RAW MATERIAL.

$(Q11 \times Q12) \times Q16$

TO DERIVE THE COST OF SALARY, WE SHOULD FIRST DETERMINE THE TIME CONSUMED ON LABOR ACTIVITIES FOR THE PRODUCTION. WE SHALL THEN MULTIPLY LABOR CONSUMED PER PRODUCT UNIT WITH PRODUCT UNIT REJECTED BY CUSTOMERS AND FURTHER MULTIPLYING THE RESULT WITH LABOR COST (PER HOUR) OF PRODUCTION.

$(Q13 \times Q11) \times Q6$

THE TOTAL COMPENSATION COST PAID TO THE CLIENTS MAY BE CALCULATED BY MULTIPLYING THE COMPENSATION COST PER FAILED UNIT BY THE NUMBER OF EXTERNAL FAILURES OCCURRED. THIS COST SHALL REFLECT UNDER "OTHER EXPENSES (D)"

$Q9 \times Q11$

5.1 EARNING FORGONE DUE TO LOST SALES

TO DETERMINE THIS WE SHALL FIRST FIND OUT THE AMOUNT OF PRODUCT UNITS WHOSE SALE HAS BEEN LOST DUE TO EXTERNAL FAILURES DURING A PERIOD OF TIME AND THIS CAN BE ESTIMATED BY MULTIPLYING THE NUMBER OF EXTERNAL FAILURES BY THE AVERAGE AMOUNT OF LOST SALES TRANSACTIONS PER EACH OF THESE FAILURES $(Q11 \times Q18)$. THEN WE SHALL DETERMINE THE PROFIT FOREGONE PER EACH OF THESE UNITS; THIS CAN BE ESTIMATED BY SUBTRACTING THE COST PER UNIT FROM THE AVERAGE PRICE $(Q8 - Q17)$. THEREFORE, THE INDIRECT COST SHALL BE DETERMINED BY THE FOLLOWING:

$(Q11 \times Q18) \times (Q8 - Q17)$

COST OF QUALITY – THE STUDY

COST OF QUALITY SUMMARY

REF. N.	COST OF QUALITY CATEGORY	AMOUNT	%
1	PREVENTION COST	8,200	1.66 %
2	APPRAISAL COST	5,208	1.05 %
3	INTERNAL FALIURE COST	141,300	28.61 %
4	EXTERNAL FALIURE COST	126,780	25.67 %
5	INDIRECT QUALITY COST	122,400	43.01 %
TOTAL COST		493,888	100.00 %

EXPLAINING THE COST OF QUALITY SUMMARY

AMOUNT COLOUMN

THE AMOUNT FOR EACH QUALITY COST CATEGORY MAY BE CALCULATED BY ADDING UP THE TOTAL COST OF THOSE SUBCATEGORIES IT COMPRISES. THESE DATA MAY BE EXTRACTED FROM THE DETAILED COST OF QUALITY RECORD.

PERCENTAGE COLOUMN

THIS CONSIST OF THE PERCENTAGE FOR EACH QUALITY COST CATEGORY MAY BE CALCULATED BY DIVIDING EACH AMOUNT BY THE TOTAL QUANTITY COST.

PREVENTION COST	EMPLOYEE TRAINING COST + PREVENTIVE MAINTENANCE COST (1.1 + 1.2)
APPRAISAL COST	RAW MATERIAL INSPECTION COST + FINAL PRODUCT INSPECTION COST (2.1 + 2.2)
INTERNAL FALIURE COST	COST OF PRODUCTS REJECTED INTERNALLY (3.1)
EXTERNAL FALIURE COST	COST OF PRODUCTS RETURNED BY CUSTOMERS (4.1)
INDIRECT QUALITY COST	EARNING FORGONE DUE TO LOST SALES (5.1)

COST OF QUALITY – THE STUDY

ACCOUNTING COST – QUALITY PERSPECTIVE

REF. N.	COST OF QUALITY CATEGORY	TOTAL COST	COST OF QUALITY	BASIC / STANDARD COST	EXPLANATION
A.	RAW MATERIAL	1,282,000	195,360	1,086,640	THIS ACCOUNT WAS AFFECTED BY THE SUBCATEGORIES 3.1 AND 4.1
B.	SALARIES / WAGES	220,000	35,928	184,072	THIS ACCOUNT WAS AFFECTED BY THE SUBCATEGORIES 2.1, 2.2, 3.1 AND 4.1
C.	OUTSOURCED SERVICE	35,000	8,200	26,800	THIS ACCOUNT WAS AFFECTED BY THE SUBCATEGORIES 1.1 AND 1.2
D.	OTHER EXPENSES	76,000	42,000	34,000	THIS ACCOUNT WAS JUST AFFECTED BY THE SUBCATEGORY 4.1
TOTAL FINANCIAL COST		1,613,000	281,488	1,331,512	
E.	INDIRECT COSTS	212,400	212,400	0	THIS ACCOUNT WAS JUST AFFECTED BY THE SUBCATEGORY 5.1
TOTAL COST		1,825,400	493,888	1,331,512	

EXPLAINING THE COST OF QUALITY SUMMARY

TOTAL COST COLOUMN
 MOST OF THE FIGURES IN "TOTAL COST" COLUMN DERIVE FROM THE FINANCIAL STATEMENTS. THE EXCEPTION IS "INDIRECT QUALITY COST" WHICH IS NOT TAKEN INTO ACCOUNT BY THE ACCOUNTING SYSTEM AND, THEREFORE, HAS TO BE ESTIMATED AFTERWARDS.

COST OF QUALITY COLOUMN
 THE FIGURES IN "QUALITY COST" COLUMN ARE OBTAINED BY ADDING UP THE AMOUNTS FOR EACH FINANCIAL ACCOUNT INCLUDED IN THE "DETAILED QUALITY COST" RECORD.

BASIC / STANDARD COST COLOUMN
 THE FIGURES IN THIS LAST COLUMN MAYBE OBTAINED BY SUBTRACTING "QUALITY COSTS" FROM "TOTAL COSTS" FOR EACH FINANCIAL ACCOUNT

- 1.1 EMPLOYEE TRAINING
- 1.2 PREVENTATIVE MAINTENANCE
- 2.1 RAW MATERIALS INSPECTION
- 2.2 FINAL PRODUCT INSPECTION
- 3.1 PRODUCTS REJECTED INTERNALLY
- 4.1 PRODUCTS REJECTED BY CUSTOMER
- 5.1 EARNINGS FORGONE DUE TO LOST SALES

VALUE STREAM MAPPING – THE STUDY

CURRENT STATE

PROCESSING TIME NOT RELATED TO MANUFACTURING		TIME (DAYS)	CLARIFICATIONS
ORDER PROCESSING TIME	WT 1	3	THE DELAY IS MAINLY DUE TO PRICE NEGOTIATION & ITEMS CLARIFICATION
WAITING TIME FOR RAW MATERIAL	WT 2	1	THE DELAY IS DUE TO SPECIAL REQUIREMENTS, 5 DAYS DELAY FOR 20% OF TOTAL ORDERS
WAITING TIME BEFORE PRODUCTION	WT 3	1	RELATED TO THE CABLE TRAY RUNS BENDING ORDERS & CABLE TRAY LINE PRODUCTIVITY
WAITING TIME BEFORE HDG	WT 4	3	RELATED TO HDG ITEMS BENDING AND HDG PLANT PRODUCTIVITY
WAITING TIME BEFORE CLEANING	WT 5	1.5	RELATED TO QUALITY ISSUES IN HDG OUTPUT
WAITING TIME BEFORE PACKAGING	WT 6	0.5	COMPILING AND SORTING TIME
WAITING TIME BEFORE DELIVERY	WT 7	2	RELATED TO AVAILABILITY OF TRANSPORTATION TO THE REQUIRED LOCATION
DELIVERY TIME	WT 8	1	RELATED TO DELIVERY LOCATION
TOTAL	WT	13	

PROCESSING TIME RELATED TO MANUFACTURING		TIME (MIN)	LABOR NO.	LABOR MIN.	CLARIFICATIONS
MANUFACTURING TIME IN CABLE TRAY LINE	PT 1	0.5	3	1.5	PRODUCTION SPEED IS 6 MTR / MIN = 120 PCS / HR
PROCESSING TIME IN HDG	PT 2	0.75	12	9	PRODUCTION AVG. FOR CABLE TRAY RUNS IS AROUND 80 PCS / HR
CLEANING TIME	PT 3	1	8	8	AVG. OUTPUT IS 60 PCS / HR BY 4 CLEANING STATIONS
PACKAGING TIME	PT 4	0.5	2	1	AVG. OUTPUT IS 240 PCS / HR BY ONE STATION (2 LABOR / STN.)
TOTAL	PT	2.7	25	19.5	

LABOR HR.0.325= LABOR MIN / 60

COST (SAR.)5.2= LABOR HR. x TOTAL HR. (2 SHIFT) = 0.325 x 16

EXAMPLEORDER QTY. LEAD TIME

TOTAL PROCESSING TIME FOR (PCS) OF CABLE TRAY *100014.04 DAYS

= WT + (QTY. x PT MAX MINS.)

= 13 + (1000 x 1 / 960)

= 14.04

CABLE TRAY MANUFACTURING TIME (VALUE ADDED TIME)19.50 MINUTES***

WAITING TIME (WASTE)13.00 DAYS

* TOTAL PROCESSING TIME

*** CONSUMED LABOR MINUTES PER PC

WT WAITING TIME

PT PROCESSING TIME

PT MAX PROCESSING TIME FOR THE SLOWEST PROCESS

◀ EFFECTED AREAS

VALUE STREAM MAPPING – THE STUDY

FUTURE STATE

PROCESSING TIME NOT RELATED TO MANUFACTURING		TIME (DAYS)	CLARIFICATIONS
ORDER PROCESSING TIME	WT 1	0.5 <	TIME REDUCED DUE TO THE IMPLEMENTATION AND UTILIZATION OF THE ERP SYSTEM
WAITING TIME FOR RAW MATERIAL	WT 2	1	THE DELAY IS DUE TO SPECIAL REQUIREMENTS, 5 DAYS DELAY FOR 20% OF TOTAL ORDERS
WAITING TIME BEFORE PRODUCTION	WT 3	1	RELATED TO THE CABLE TRAY RUNS BENDING ORDERS & CABLE TRAY LINE PRODUCTIVITY
WAITING TIME BEFORE HDG	WT 4	0.5 <	TIME REDUCED DUE TO THE INCREASE IN HDG CAPACITY
WAITING TIME BEFORE CLEANING	WT 5	0.5 <	TIME REDUCED DUE TO THE INCREASE IN HDG OUTPUT QUALITY
WAITING TIME BEFORE PACKAGING	WT 6	0.5	COMPILING AND SORTING TIME
WAITING TIME BEFORE DELIVERY	WT 7	1 <	TIME REDUCED DUE TO BETTER MANAGEMENT USING ERP SYSTEM
DELIVERY TIME	WT 8	1	RELATED TO DELIVERY LOCATION
TOTAL	WT	6 <	

PROCESSING TIME RELATED TO MANUFACTURING		TIME (MIN)	LABOR NO.	LABOR MIN.	CLARIFICATIONS
MANUFACTURING TIME IN CABLE TRAY LINE	PT 1	0.5	3	1.5	PRODUCTION SPEED IS 6 MTR / MIN = 120 PCS / HR
PROCESSING TIME IN HDG	PT 2	0.4 <	10	4	PRODUCTION AVG. FOR CABLE TRAY RUNS IS AROUND 150 PCS / HR
CLEANING TIME	PT 3	0.5 <	8	4	AVG. OUTPUT IS 120 PCS / HR BY 4 CLEANING STATIONS
PACKAGING TIME	PT 4	0.5	2	1	AVG. OUTPUT IS 240 PCS / HR BY ONE STATION (2 LABOR / STN.)
TOTAL	PT	1.9 <	23	10.5	

LABOR HR.0.175= LABOR MIN / 60

COST (SAR.)2.8= LABOR HR. x TOTAL HR. (2 SHIFT) = 0.175 x 16

EXAMPLEORDER QTY. LEAD TIME

TOTAL PROCESSING TIME FOR (PCS) OF CABLE TRAY *10006.52 DAYS

= WT + (QTY. x PT MAX MINS.)

= 6 + (1000 x 0.5 / 960)

= 6.52

CABLE TRAY MANUFACTURING TIME (VALUE ADDED TIME)10.50 MINUTES***

WAITING TIME (WASTE)6.00 DAYS

* TOTAL PROCESSING TIME

*** CONSUMED LABOR MINUTES PER PC

WT WAITING TIME

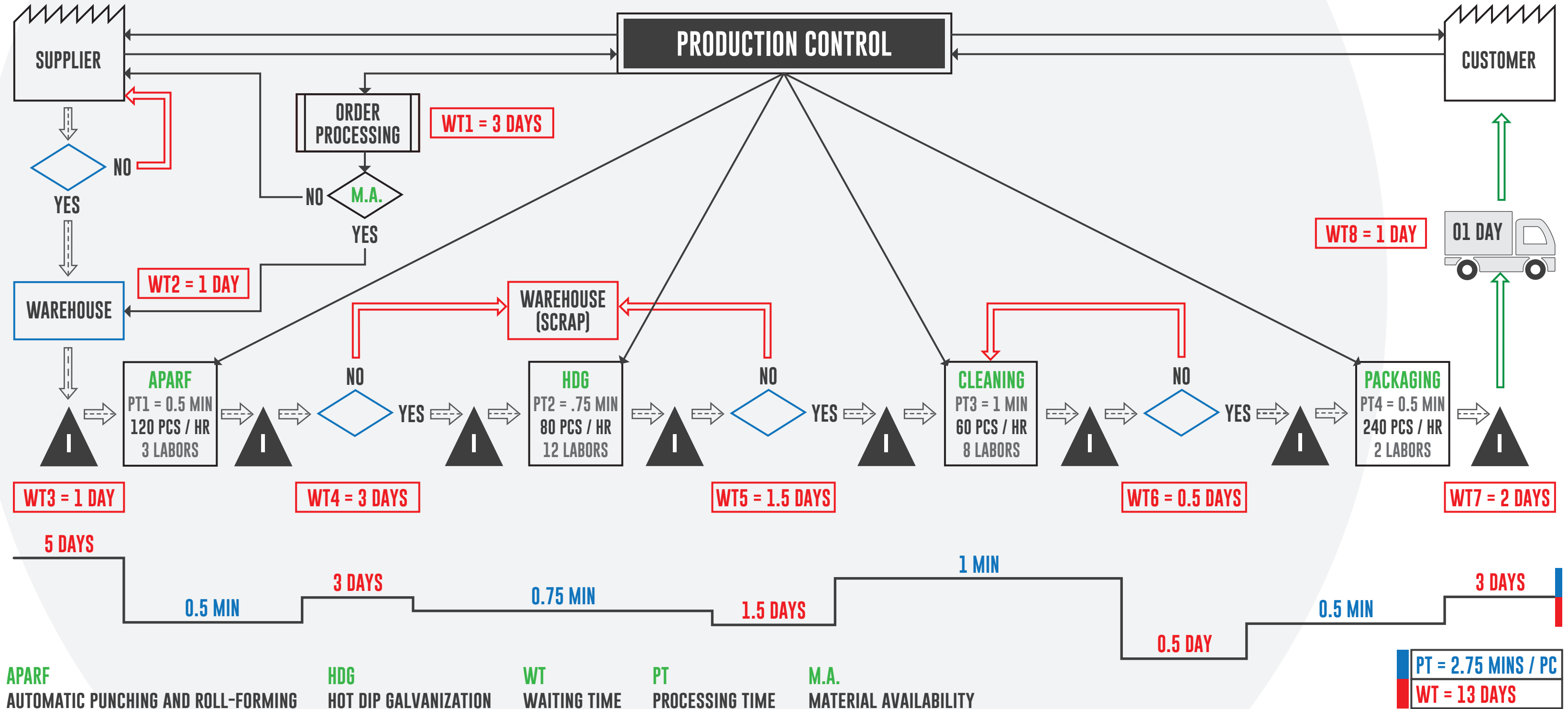
PT PROCESSING TIME

PT MAX PROCESSING TIME FOR THE SLOWEST PROCESS

< EFFECTED AREAS

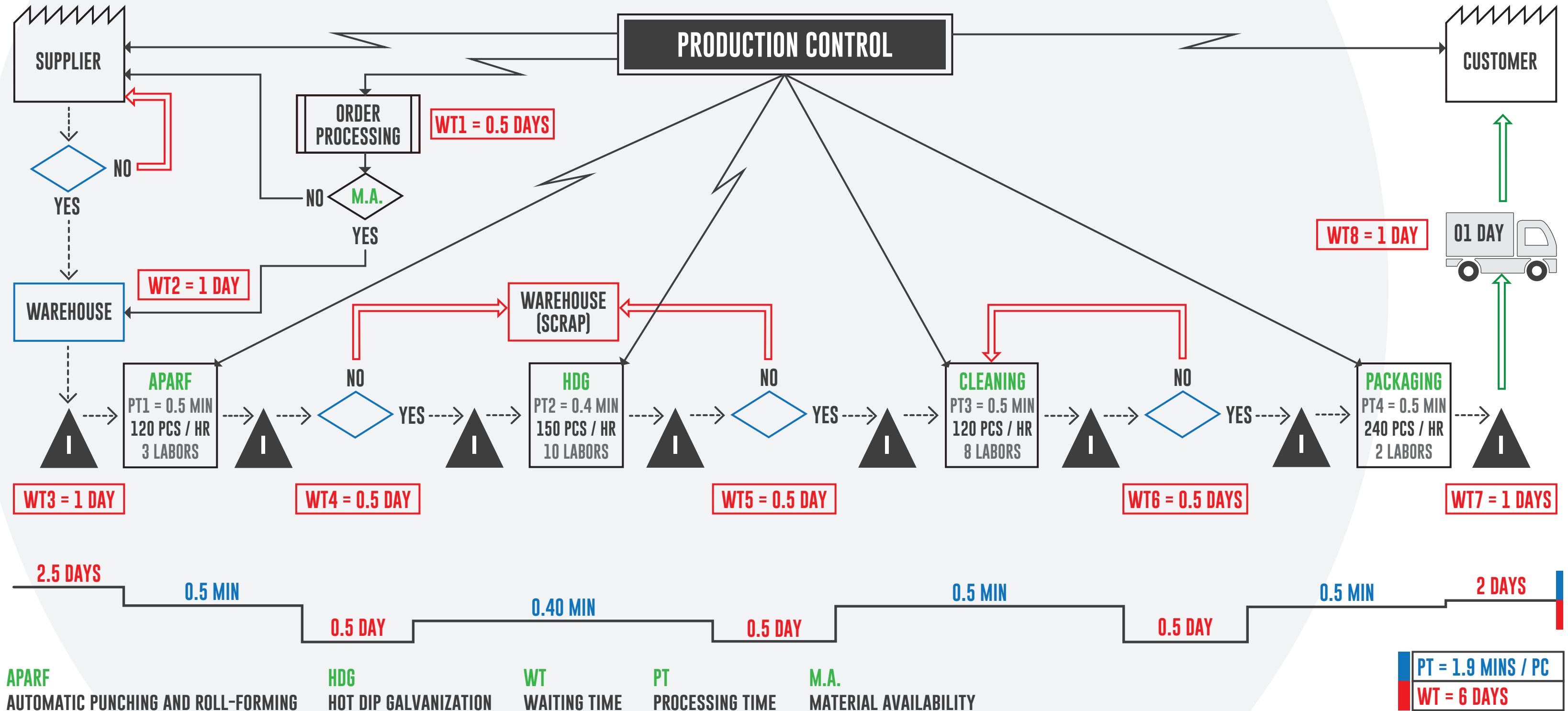
VALUE STREAM MAPING

CURRENT STATE



VALUE STREAM MAPING

FUTURE STATE



VALUE STREAM MAPPING

COST – BENEFIT ANALYSIS

INVESTMENT IN IMPROVEMENT	CURRENT VALUE (SAR)	INVESTMENT (SAR)	NET INVESTMENT (SAR)
MANAGEMENT SOFTWARE (ERP)	-	750,000	750,000
HDG PLANT (6.5 X 1.5 X 3.5 MTR.)	250,000	8,500,000	8,250,000
TOTAL			9,000,000

	UNIT	CURRENT STATE	FUTURE STATE	COMMENTS
CABLE TRAY RUNS CAPACITY PER HOUR	PCS / HOUR	60.00	120.00	BASED ON MAXIMUM PROCESSING TIME (PT MAX) (BOTTLENECK)
CABLE TRAY RUNS CAPACITY PER MONTH	PCS / MONTH	24,960.00	49,920.00	BASED ON 16 WORKING HOURS PER DAY & 26 DAYS PER MONTH
MANUFACTURING TIME PT	MIN / PC	2.75	1.90	TOTAL MANUFACTURING TIME OF ALL PROCESSES FOR EACH PC
CABLE TRAY MANUFACTURING TIME (VALUE ADDED TIME)	LABOR MIN.	19.50	10.50	CONSUMED LABOR MINUTES PER PC
LABOR MIN. AVERAGE COST	SAR	0.27	0.27	BASED ON 16 SAR / HOUR
CABLE TRAY RUNS LABOR COST	SAR / PC	5.20	2.80	46% LABOR COST REDUCTION PER PC
CABLE TRAY RUNS WAITING TIME	DAY	13.00	6.00	WASTED TIME
AVERAGE MONTHLY PRODUCTION	PCS	23,500.00	46,000.00	BASED ON MARKET DEMAND AND DUE TO IMPROVED DELIVERY TIME
AVERAGE MONTHLY LABOR COST FOR CABLE TRAY RUNS	SAR	122,200.00	128,800.00	
RAW MATERIAL COST	SAR / PC	60.00	60.00	RAW AND HDG COST (4 SAR / KG X AVG. WEIGHT 15 KG / PC)
SELLING PRICE	SAR / PC	105.00	105.00	BASED ON 7 SAR / KG AS SELLING PRICE
GROSS PROFIT	SAR / PC	45.00	45.00	
MONTHLY GROSS PROFIT	SAR / MONTH	1,057,500.00	2,070,000.00	
CALCULATED MONTHLY NET COST - BENEFIT	SAR		1,005,900.00	= GROSS PROFIT DIFFERENCE + LABOR COST DIFFERENCE

**"WHAT IS THE DIFFERENCE BETWEEN
AN OBSTACLE AND AN OPPORTUNITY?"**

OUR ATTITUDE TOWARDS IT

**EVERY OPPORTUNITY HAS A DIFFICULTY, AND
EVERY DIFFICULTY HAS AN OPPORTUNITY."**

***Lets Challenge !
The times to come***

